NITHIN KIDANGAZHIATH MANA

nkidang1@jh.edu | +14105001549 | My Website | My LinkedIn

SUMMARY

Digital Design and RTL-focused ECE graduate from Johns Hopkins University with experience in AI hardware acceleration, FPGA and ASIC design, and quantized neural network architectures. Skilled in Verilog and SystemVerilog languages with demonstrated work in Vision Mamba and in-memory computing. Seeking to contribute to the design and development of scalable, high-performance digital systems and next-generation computing architectures.

EDUCATION

JOHNS HOPKINS UNIVERSITY | Baltimore, MD
MSE in Electrical and Computer Engineering | Graduated - 05/2026 | 3.6/4.0

VELLORE INSTITUTE OF TECHNOLOGY, VIT | Tamil Nadu, India
B.Tech in Electrical and Electronics Engineering |Graduated - 07/2024 | 3.7/4.0

TECHNICAL PROJECTS

Investigating State-Space ML Models for Memory-Efficient 3D Brain Tumor Segmentation (Submission to NeurIPS 2026) | Johns Hopkins University | Baltimore, MD | January 2026 – May 2026 |

- Replaced shifted-window Transformer encoder blocks in **Swin-UNETR** with **Vision Mamba (SSM) modules** to evaluate state-space models as scalable alternatives for **3D medical image segmentation using PyTorch/Python** (1).
- Achieved 20% reduction in **peak GPU training memory,** 46.6% reduction in **peak inference memory, and comparable Dice and Hausdorff Distance** scores relative to Swin-UNETR baseline.

Design of 8-Bit & Binary Neural Network Accelerators| Johns Hopkins University | Baltimore, MD | August - December 2024

- Performed **Verilog RTL Design using Cadence Xcelium of** 8-bit quantized and binarized **neural network ASIC accelerators** for image classification, demonstrating an ~8% reduction in combinational logic in the BNN design with equal area.
- Analyzed power distribution across combinational, sequential, memory, and IO components using post-layout PPA reports on 45nm technology.

CERTIFICATIONS

- Cadence SystemVerilog for Design and Verification v25.03 Certification (Certified Feb 2026)
- Cadence SystemVerilog Assertions v5.2 Certification (Certified March 2026)
- Cadence SystemVerilog Accelerated Verification with UVM v1.2.5rev2 **(Anticipated Certification May 2026)**

WORK EXPERIENCE

Teaching Assistant: First-Year ECE Design | Johns Hopkins University | Baltimore, MD | January 2026 – May 2026

- Mentored 30+ students over weekly lab sections and office hours, teaching circuit fundamentals and core electrical design concepts.
- Graded homework, quizzes, lab reports, and communicated technical feedback.

ML Hardware Architect - Research| Johns Hopkins University | Baltimore, MD | May 2025 - January 2026

- Led **Verilog RTL Design** of 8-bit fixed-point quantized Vision Mamba **FPGA accelerator** for MedMNIST classification (2).
- Delivered **9 times reduction in latency** (compared to baseline) via channel-level architecture parallelization.
- Collaborated with doctoral team members for implementation and verification efforts to meet hardware and resource utilization constraints for **Kintex-7 XC7K325T FPGA,** using **AMD Vivado 2024.1**.
- **Published in IEEE Transactions on Biomedical Circuits and Systems**

Technical R&D Intern | STMicroelectronics | Uttar Pradesh, India | January 2024 - June 2024

- Spearheaded **RTL Design** of a 6T SRAM-based Analog In-Memory Compute architecture using **Simulink-based Verilog auto-generation** and **Cadence Xcelium simulation tool**.
- Developed a noise-aware hardware validation framework based on LFSRs to model ±0.2% bit-cell noise and ±1 feature-level perturbations, ensuring robustness in mixed-signal operation.

VLSI Design Intern | Maven Silicon Pvt. Ltd | Karnataka, India | May 2023 - July 2023

- Conducted **Verilog RTL design** for an **AMBA AHB-to-APB bridge** supporting INCR4 and WRAP4 burst transfers, address pipelining, and FSM-controlled APB transfers using **Intel Quartus Prime software**.